

National University of Computer and Emerging Sciences, Lahore Campus				
	Course:	Machine Learning for Robotics	Course Code:	CS 5096
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Date:	09-02-2026	Total Marks:	20
	Section:	A	Weight:	
	Exam:	Quiz 1	Roll No:	

Q1) Use a single iteration of gradient descent to update the weights of a linear regression model on the following dataset. Let $\theta_0 = 0.5$, $\theta_1 = 0.5$, $\alpha = 0.1$ initially. The cost function is given by: (13)

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Find the updated values of θ_0 and θ_1 after this iteration.

Driving Experience (years)	Monthly Auto Insurance Premium (\$)
5	64
2	87
12	50

Q2) Use the normal equation to find the exact weights for the dataset in Q1. (7)

$$\text{Let } X = \begin{bmatrix} 1 & 5 \\ 1 & 2 \\ 1 & 12 \end{bmatrix}, y = \begin{bmatrix} 64 \\ 87 \\ 50 \end{bmatrix}, \text{ and } X^T X = \begin{bmatrix} 3 & 19 \\ 19 & 173 \end{bmatrix}.$$